

LAB EXPERIMENT RECORD

Set 6: Product Sales Analysis

Department	CSE-Ds
Course Code / Course Name	DSA0601 / Data Handling and Visualization
Experiment	Data Visualization using R
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Aim

To analyze and visualize monthly product sales data using a grouped bar chart, stacked area chart, and table, and to create an interactive dashboard for exploring product sales data.

Dataset: Monthly Product Sales

Product ID	Product Name	January Sales	February Sales	March Sales
1	Product A	2000	2200	2400
2	Product B	1500	1800	1600
3	Product C	1200	1400	1100

PROGRAM 1: Grouped Bar Chart of Product Sales

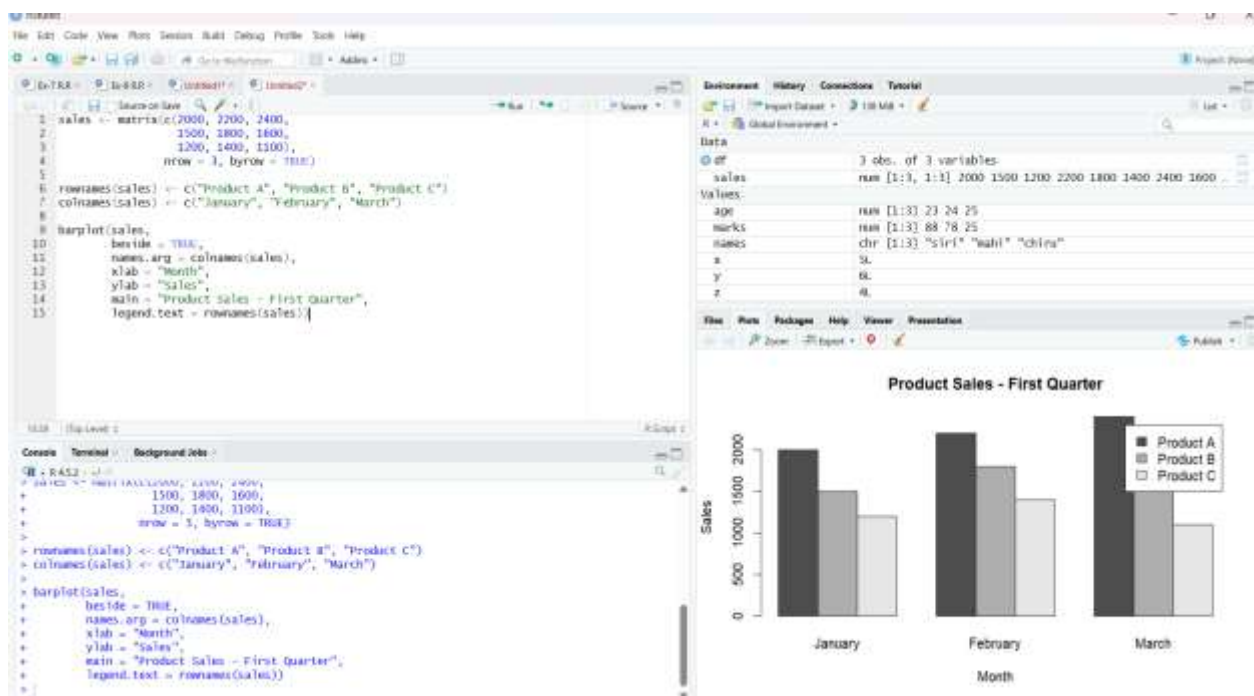
Create a grouped bar chart to visualize the sales of each product for the first quarter. Label the chart elements.

R Program

R

```
sales <- matrix(c(2000, 2200, 2400,  
                1500, 1800, 1600,  
                1200, 1400, 1100),  
              nrow = 3, byrow = TRUE)  
rownames(sales) <- c("Product A", "Product B", "Product C")  
colnames(sales) <- c("January", "February", "March")  
barplot(sales,  
        beside = TRUE,  
        names.arg = colnames(sales),  
        xlab = "Month",  
        ylab = "Sales",  
        main = "Product Sales - First Quarter",  
        legend.text = rownames(sales))
```

Output 1: Product Sales Grouped Bar Chart



Observation

The grouped bar chart compares the monthly sales of Product A, Product B, and Product C during the first quarter. Product A has the highest sales in all three months, while Product C has the lowest sales. Product A also shows a continuous increase from January to March.

PROGRAM 2: Stacked Area Chart of Product Sales

Generate a stacked area chart to represent the overall sales trend for all products over the three months.

R Program

R

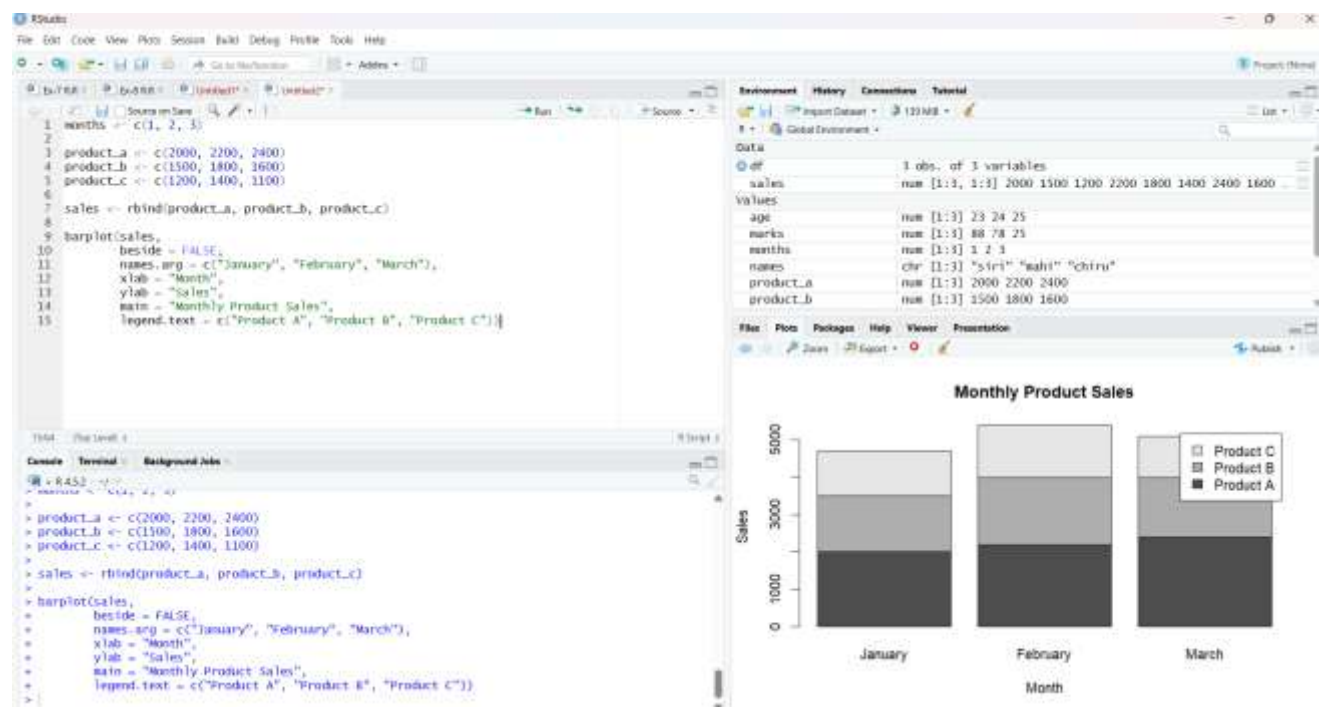
```
months <- c("January", "February", "March")

product_a <- c(2000, 2200, 2400)
product_b <- c(1500, 1800, 1600)
product_c <- c(1200, 1400, 1100)
sales <- rbind(product_a, product_b, product_c)

matplot(months,
        t(sales),
        type = "l",
        lty = 1,
        lwd = 2,
        xlab = "Month",
        ylab = "Sales",
        main = "Monthly Product Sales")

legend("topleft",
        legend = c("Product A", "Product B", "Product C"),
        lty = 1,
        lwd = 2)
```

Output 2: Product Sales Stacked Area Chart



Observation

The chart represents the sales trend of all three products over the three months. Product A shows a continuous increase in sales, while Product B increases in February and decreases in March. Product C increases in February and decreases in March.

PROGRAM 3: Table of Monthly Product Sales

Build a table to show the monthly sales data for each product. Label the table elements.

R Program

R

```
product_id <- c(1, 2, 3)
```

```
product_name <- c("Product A", "Product B", "Product C")
```

```
january <- c(2000, 1500, 1200)
```

```
february <- c(2200, 1800, 1400)
```

```
march <- c(2400, 1600, 1100)
```

```
sales_data <- data.frame(
```

```
  Product_ID = product_id,
```

```
  Product_Name = product_name,
```

```

January_Sales = january,

February_Sales = february,

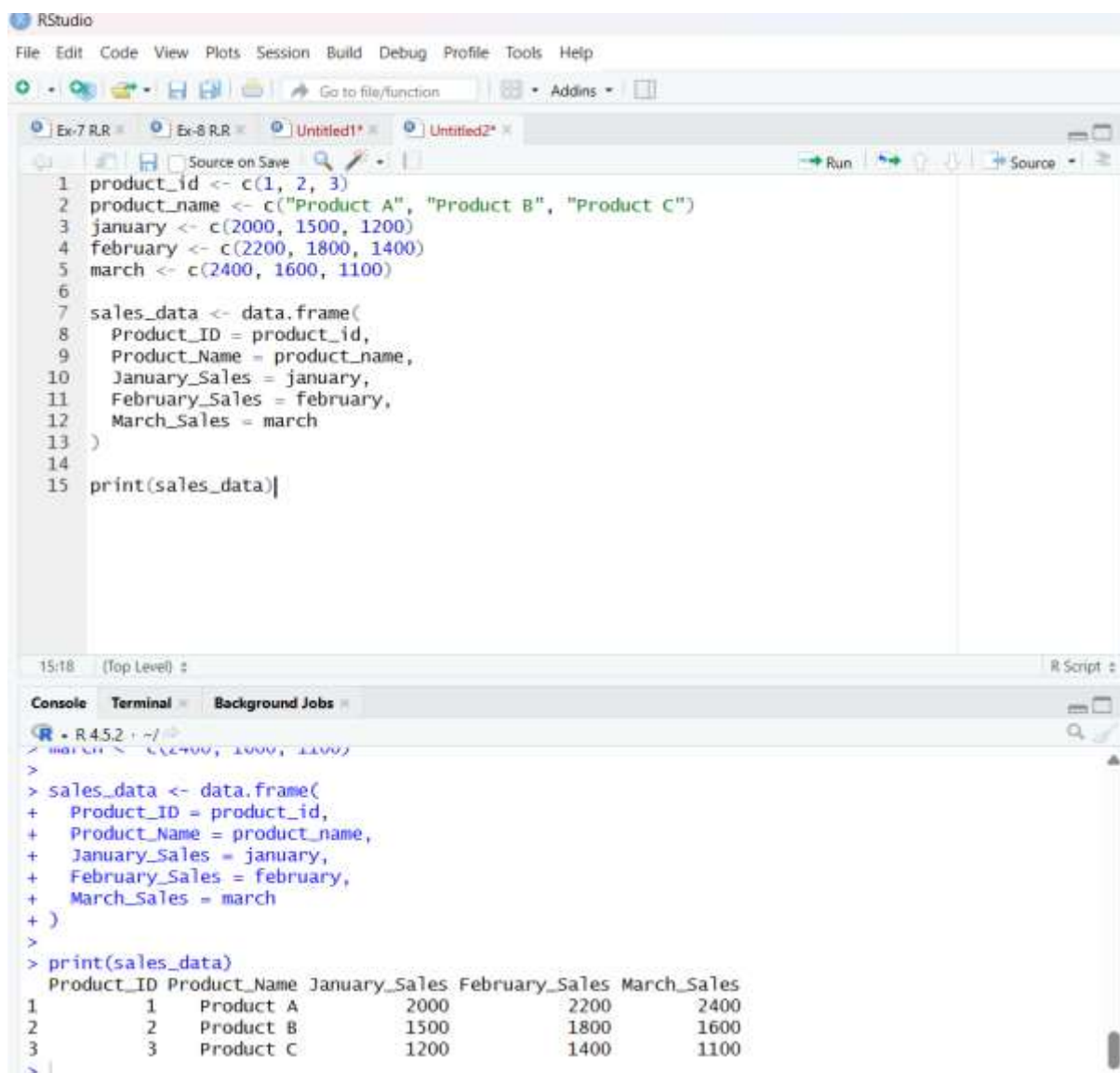
March_Sales = march

)

print(sales_data)

```

Output 3: Monthly Product Sales Table



The screenshot shows the RStudio interface. The script editor contains the following R code:

```

1 product_id <- c(1, 2, 3)
2 product_name <- c("Product A", "Product B", "Product C")
3 january <- c(2000, 1500, 1200)
4 february <- c(2200, 1800, 1400)
5 march <- c(2400, 1600, 1100)
6
7 sales_data <- data.frame(
8   Product_ID = product_id,
9   Product_Name = product_name,
10  January_Sales = january,
11  February_Sales = february,
12  March_Sales = march
13 )
14
15 print(sales_data)

```

The console output shows the execution of the code and the resulting data frame:

```

> sales_data <- data.frame(
+   Product_ID = product_id,
+   Product_Name = product_name,
+   January_Sales = january,
+   February_Sales = february,
+   March_Sales = march
+ )
> print(sales_data)
  Product_ID Product_Name January_Sales February_Sales March_Sales
1          1   Product A         2000         2200         2400
2          2   Product B         1500         1800         1600
3          3   Product C         1200         1400         1100

```

Observation

The table displays the monthly sales values of all three products. It provides a clear comparison of January, February, and March sales for each product.

PROGRAM 4: Interactive Product Sales Dashboard

Develop a Tableau dashboard combining the grouped bar chart, stacked area chart, and the table for interactive exploration of sales data.

Data

Product ID	Product Name	January Sales	February Sales	March Sales
1	Product A	2000	2200	2400
2	Product B	1500	1800	1600
3	Product C	1200	140	1100

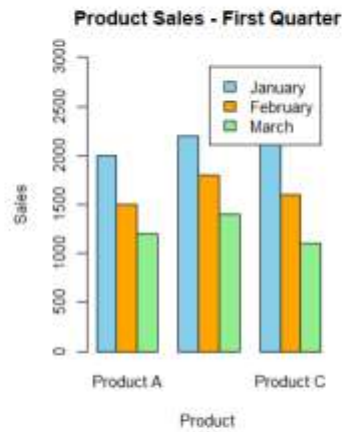
Output 4: Interactive Product Sales Dashboard

Product Sales Analysis Dashboard

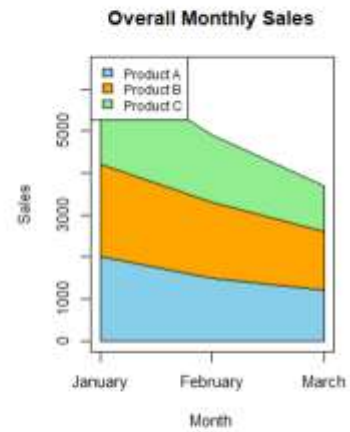
Select Product:

All

Grouped Bar Chart



Stacked Area Chart



Monthly Product Sales

Product	January	February	March
Product A	2000.00	2200.00	2400.00
Product B	1500.00	1800.00	1600.00
Product C	1200.00	1400.00	1100.00

Observation

The dashboard combines the grouped bar chart, stacked area chart, and sales table in a single view. It helps users compare product performance, observe monthly sales trends, and view exact sales values interactively.